CARBON EXCHANGE REGISTRY AND SETTLEMENT ENGINE

Software Design Document – Week 1

Document field	Details
Domain	Energy – Carbon Markets
Technology	Python 3.10+
Application type	Command-file-driven backend application; no UI
Checkpoint	Week 1 – Design document
Version	1.0
Prepared for	Capstone project team

Deliverables: Glossary • UML Class Diagram • Certificate State Diagram • Business-Rule Register

1. Introduction

1.1 Purpose

This document defines the Week 1 design for a registry and settlement engine used by a carbon exchange. The system manages Renewable Energy Certificates (RECs) and carbon credits. It supports generator and obligated-entity registration, certificate issuance, double-auction trading, certificate transfers, financial settlement, RPO retirement, expiry, reversals and compliance reporting.

1.2 Scope

The system will:

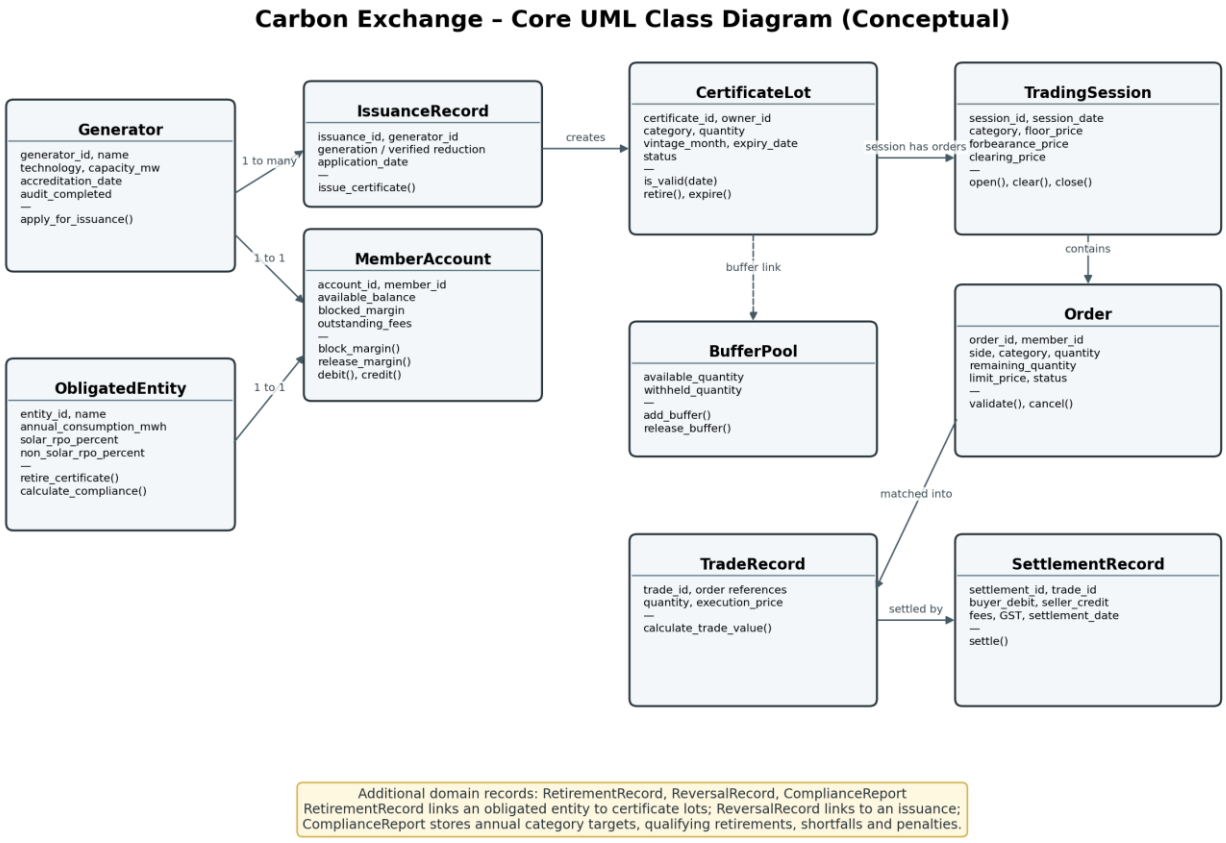
- Register renewable generators using solar, wind or biomass technology and register obligated entities.
- Issue RECs from metered generation and carbon credits from verified emission reductions.
- Track certificate category, ownership, vintage, quantity and expiry.
- Accept buy and sell orders and clear periodic double-auction sessions at a uniform price.
- Transfer eligible certificates FIFO by vintage and maintain member account balances and margins.
- Retire certificates against solar and non-solar RPO targets and calculate shortfall penalties.
- Process reversals, buffer-pool recovery, certificate expiry and registry/trading/compliance reports.
- Read commands from files and produce output without a graphical user interface.

2. Glossary of Terms

Term	Definition
REC	Renewable Energy Certificate representing 1 MWh of renewable electricity generation.
Carbon Credit	A unit representing one tonne of verified CO2-equivalent emission reduction, subject to buffer withholding.
Generator	An entity producing renewable electricity through solar, wind or biomass technology.
Obligated Entity	An organization required to meet prescribed renewable purchase obligations.
RPO	Renewable Purchase Obligation; the required renewable-energy procurement percentage.
Solar REC	REC generated from solar electricity.
Non-Solar REC	REC generated from non-solar renewable sources such as wind or biomass.
Vintage	The generation or issuance month associated with a certificate.
Expiry Date	The last date on which a certificate remains valid for trading and retirement.
Double Auction	A market mechanism in which buyers submit bids and sellers submit offers.
Buy Order	An instruction to purchase certificates at a specified maximum price.
Sell Order	An instruction to sell certificates at a specified minimum price.
Clearing Price	The single uniform price at which matched orders are settled.
Floor Price	The minimum permitted REC trading price: Rs 1,000.
Forbearance Price	The maximum permitted REC trading price: Rs 3,000.
Marginal Price	The price level at which remaining eligible supply or demand is allocated proportionally.
Pro-Rata Allocation	Distribution of available quantities proportionally among orders at the marginal price.
FIFO	First-In, First-Out; oldest eligible certificate vintages are transferred first.
Buffer Pool	A reserve of withheld carbon credits used to cover potential reversals.
Retirement	Permanent removal of a certificate from circulation to satisfy an RPO obligation.
Reversal	Recovery or cancellation of previously issued credits due to verification failure.
Margin	Funds blocked against a buy order to support payment.
Exchange Fee	A transaction fee charged by the exchange on trade value.
Registry Fee	A certificate-transfer fee charged to the seller.
GST	Goods and Services Tax applied to applicable fees at 18%.
Settlement	Financial completion of a trade, including fees and proceeds.
T+1	Settlement of fees on the next business day after the trade date.
Shortfall	The difference between required RPO units and qualifying retired units.
Carry-Forward	Permission for an eligible RPO shortfall to be carried into the next compliance year once.
Metered Generation	Actual renewable electricity generation recorded by an approved meter.
Verified Emission Reduction	An independently verified reduction in greenhouse-gas emissions.
Accreditation	Official recognition of a generator's eligibility for certificate issuance.

3. UML Class Diagram

The conceptual class diagram identifies the principal domain objects and their relationships. Implementation should use quantity-based certificate lots so partial transfers, reservations, retirements, reversals and expiry can be tracked independently.



3.1 Main class responsibilities

Class	Key attributes	Responsibility
Generator	generator_id, name, technology, capacity_mw, accreditation_date, audit_completed	Registration and issuance eligibility.
ObligatedEntity	entity_id, name, annual_consumption_mwh, solar_rpo_percent, non_solar_rpo_percent	Certificate retirement and RPO compliance.
MemberAccount	account_id, member_id, available_balance, blocked_margin, outstanding_fees	Margin blocking, release, debit and credit.
IssuanceRecord	issuance_id, generator_id, generation/verified reduction, application_date	Records the basis and approval of an issuance.
CertificateLot	certificate_id, owner_id, category, quantity, vintage_month, expiry_date, status	Tracks certificate quantity, ownership and lifecycle.
BufferPool	available_quantity, withheld_quantity	Tracks withheld credits and eligible releases.
Order	order_id, member_id, side, category, quantity, remaining_quantity, limit_price, status	Stores and validates buy/sell orders.
TradingSession	session_id, date, category, floor_price, forbearance_price, clearing_price	Controls auction opening, clearing and closing.
TradeRecord	trade_id, buy_order_id, sell_order_id, quantity, execution_price	Records each matched trade.
SettlementRecord	settlement_id, trade_id, buyer_debit, seller_credit, fees, GST	Records financial settlement.
RetirementRecord	retirement_id, entity_id, certificate_id, quantity, retirement_date	Records compliance retirements.
ReversalRecord	reversal_id, issuance_id, reversed_quantity, outstanding_liability	Records reversal recovery and deficit.
ComplianceReport	entity_id, compliance_year, targets, retired quantities, penalty	Stores annual compliance results.

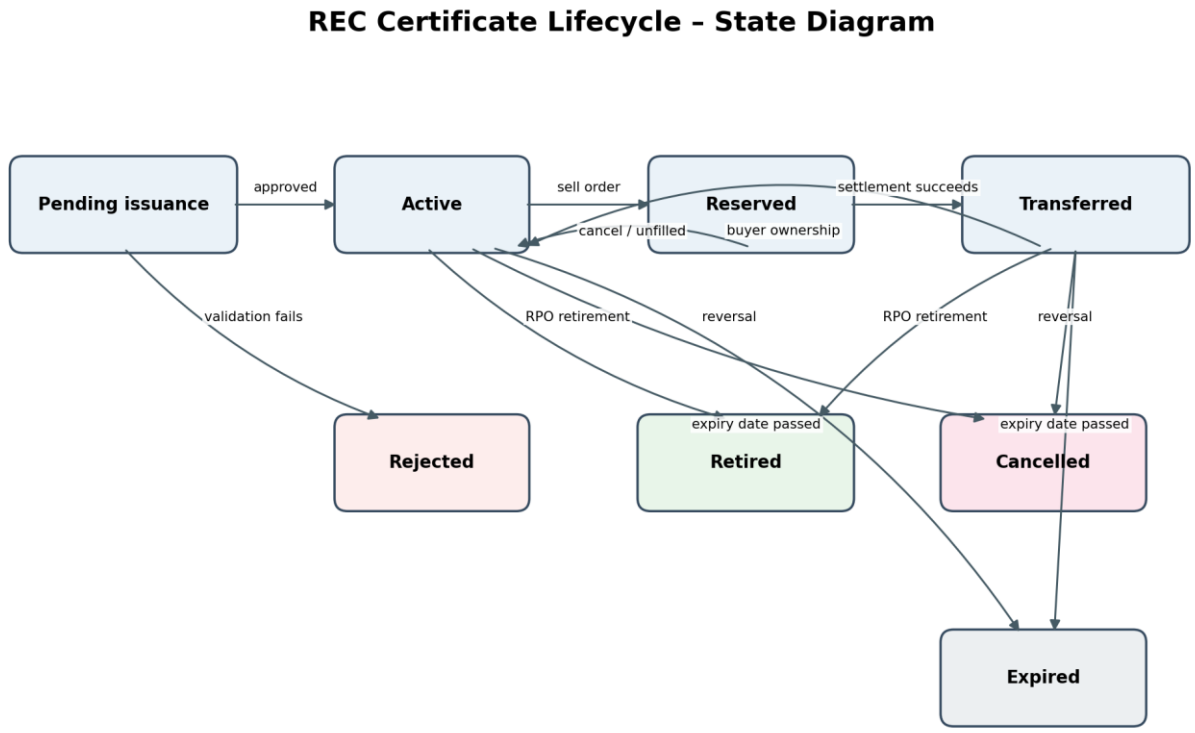
3.2 Relationship summary

- One generator can have many issuance records; an issuance can create multiple certificate lots.
- Each generator and obligated entity has a member account.

- A trading session contains multiple orders and can produce multiple trade records.
- Each executed trade has a settlement record.
- An obligated entity can have multiple retirement records and annual compliance reports.
- A reversal record references the issuance being reversed; buffer-pool movements are recorded against the relevant issuance.

4. UML State Diagram – REC Certificate Lifecycle

The state diagram models the lifecycle of a certificate quantity or lot, from application through issuance and possible trading, retirement, cancellation or expiry.



Terminal states: Rejected, Retired, Cancelled and Expired. State changes apply to certificate quantities/lots; partial transfers or retirements may split a lot into separately tracked quantities.

State	Description
PendingIssuance	Application submitted; validation and approval pending.
Rejected	Issuance application failed validation.
Active	Issued certificate quantity available for eligible use.
Reserved	Quantity temporarily encumbered by an accepted sell order.
Transferred	Ownership changed after successful trade settlement.
Retired	Quantity permanently used for RPO compliance.
Cancelled	Quantity cancelled due to reversal or invalidation.
Expired	Validity period has ended.

4.1 State-transition rules

- A certificate becomes Active only after successful issuance validation.
- An Active quantity may become Reserved when a valid sell order is accepted.
- Cancelled or unfilled sell orders release the reserved quantity back to Active holdings.
- Successful settlement transfers ownership to the buyer; the buyer's received lot is available subject to category and expiry rules.
- Eligible Active or Transferred quantities may be retired against RPO obligations.
- Expired quantities cannot be traded or retired. Expiry is inclusive: trading on the expiry date is valid.

- Retired, Cancelled and Expired quantities cannot return to active circulation.

Note: A partial transfer or retirement may split one original lot into multiple quantity portions. The state is therefore tracked at the lot/quantity-portion level, not only at the original issuance level.

5. Business-Rule Register

Each rule has a unique identifier for traceability between the design, implementation and test cases.

Rule ID	Rule name	Business rule / expected behaviour
BR-01	REC issuance ratio and application deadline	Issue one REC per 1 MWh of metered generation, rounding down. The application must be submitted within six calendar months of the generation month.
BR-02	Generator accreditation	A generator accredited for less than one year may receive issuance only after its audit flag is set.
BR-03	Carbon credit issuance	Issue one carbon credit per verified tonne of CO2e reduction. Withhold 5% of each issuance into the buffer pool.
BR-04	Buffer pool release	Release buffer credits to the generator after three consecutive issuances without reversal.
BR-05	Certificate category segregation	Solar and non-solar RECs are separate markets and targets. Solar RECs may satisfy non-solar shortfall only after the solar target is fully met; non-solar RECs cannot satisfy solar shortfall.
BR-06	Vintage and expiry	RECs expire 36 months after the end of the issuance month. Expired certificates cannot be traded or retired. A trade matched on the expiry date is valid.
BR-07	Auction clearing	Sort buys by descending price and sells by ascending price. Clear at a single uniform price.
BR-08	Marginal pro-rata allocation	Allocate marginal-price orders proportionally by quantity, round down to whole units and leave residual units unfilled.
BR-09	Floor and forbearance constraints	Reject REC order prices outside Rs 1,000–Rs 3,000. If the calculated clearing price exceeds forbearance, eligible sells at or below forbearance clear and buys are pro-rated.
BR-10	Fees and GST	Charge exchange fee of 0.5% of trade value to both sides and registry transfer fee of Rs 2 per certificate to the seller. Apply GST of 18% to fees. Fees settle T+1.
BR-11	Buy-order margin	Block 100% of buy-order value at entry. Release margin for rejected or unfilled quantities and convert the filled portion to settlement.
BR-12	FIFO certificate transfer	Transfer certificates using First-In, First-Out ordering by vintage.
BR-13	RPO targets and retirement	Solar target is 10% and non-solar target is 15% of annual consumption. Retirements before 31 March count toward the compliance year.
BR-14	RPO carry-forward	An entity retiring at least 90% of required units may carry the remaining shortfall to the next year once.
BR-15	Reversal handling	Cancel untraded generator units first, then buffer-pool units. Any remaining deficit becomes a monetary liability at the last clearing price. Already-retired units are not clawed back.
BR-16	Order validation and cancellation	Sell quantity cannot exceed unencumbered holdings of the same category. Partial sells are allowed. Orders cannot be modified; cancellation after session close is rejected.
BR-17	Expiry processing	Exclude expired certificates from eligible trading and retirement holdings and reflect expiry in registry reports.

5.1 Important interpretations for implementation

Rule	Implementation interpretation
BR-01	Use calendar-month arithmetic, not a fixed 180-day period. For March generation, September 30 is the six-month deadline under the stated interpretation.
BR-04	Track consecutive successful issuances per generator; a reversal resets the applicable consecutive-issuance count. Confirm reset semantics against evaluator guidance if available.
BR-06	Treat expiry as inclusive: the certificate remains valid through its expiry date.
BR-07	When equal executable volume is achieved at two prices, choose the lower price.
BR-08	Do not distribute residual units using an undocumented rounding adjustment.
BR-11	Check total settlement affordability, including buyer-side fees and GST, so the account does not become unintentionally negative.
BR-14	Track carry-forward eligibility by entity, category and compliance year to prevent repeated use.
BR-15	Record recovered quantities, buffer movements, last applicable clearing price and outstanding monetary liability.

6. Design Assumptions and Open Questions

ID	Assumption / open question
DA-01	Carbon credits are treated as a separate market unless the evaluator specifies that they share REC auctions.
DA-02	Each trading session is category-specific.
DA-03	Price discovery should maximize executable volume, minimize imbalance, and choose the lower price when otherwise tied; validate against visible expected outputs.
DA-04	Use Decimal for monetary arithmetic and document a two-decimal currency rounding policy, subject to sample-output confirmation.
DA-05	For an issuance month of March 2026, expiry is March 31, 2029, with validity through that date.
DA-06	Represent certificates as quantity-based lots to support partial transfers and independent lifecycle states.
DA-07	Settlement must be atomic: failed settlement must not leave certificate ownership changed without corresponding financial settlement.

DA-08	Track carry-forward by entity, category and year; clarify how carried units affect the following year's target.
DA-09	The whole-credit rounding convention for the 5% carbon buffer must be confirmed against sample data or evaluator guidance.
DA-10	Buy-order margin is 100% of order value; buyer fees must also be funded without creating a negative available balance.

The problem statement does not fully define carbon-credit auction treatment, buffer rounding, all tie-breaking details for price discovery, or the accounting treatment of carried-forward units. These items should be confirmed or explicitly documented before implementation.

7. Conclusion

This design establishes the foundation for the Carbon Exchange Registry and Settlement Engine. The glossary defines domain terminology; the class diagram describes the principal objects and relationships; the state diagram models the REC lifecycle; and the business-rule register provides traceability for all 17 identified rules. These design artefacts will guide implementation and testing in the subsequent project weeks.